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EXPERIMENT 8

Aim: Association Rule Mining using Apriori Algorithm.

Example:

Sure! Let's replace the food items with **tech gadgets** to keep it interesting. Here's the updated example using the same logic but with **new item names**:

New Transaction Dataset (Tech Items)

Transaction ID	Items Purchased
1	Laptop, Smartphone, Headphones
2	Laptop, Smartphone
3	Laptop, Smartwatch
4	Smartphone, Headphones
5	Laptop, Smartphone, Headphones, Smartwatch

Step 1: Find Frequent Item Set and Their Support

Item	Frequency	Support (%)
Laptop	4	$4/5 = 80\%$
Smartphone	4	$4/5 = 80\%$
Headphones	3	$3/5 = 60\%$
Smartwatch	2	$2/5 = 40\%$

Remove items with support < 50%

Remaining items: Laptop, Smartphone, Headphones

Step 2: 2-item Candidate Sets

Item Pair	Frequency	Support (%)
Laptop, Smartphone	3	$3/5 = 60\%$
Laptop, Headphones	2	$2/5 = 40\%$
Smartphone, Headphones	3	$3/5 = 60\%$

Step 3: Generate Association Rules



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A) Rule: Laptop → Smartphone

- Confidence = $(3/5) \div (4/5) = 75\%$

B) Rule: Smartphone → Laptop

- Confidence = $(3/5) \div (4/5) = 75\%$

C) Rule: Smartphone → Headphones

- Confidence = $(3/5) \div (4/5) = 75\%$

D) Rule: Headphones → Smartphone

- Confidence = $(3/5) \div (3/5) = 100\%$

Final Valid Rules (Support $\geq 50\%$, Confidence $\geq 75\%$)

1. **Laptop → Smartphone** (Support: 60%, Confidence: 75%)
2. **Smartphone → Laptop** (Support: 60%, Confidence: 75%)
3. **Smartphone → Headphones** (Support: 60%, Confidence: 75%)
4. **Headphones → Smartphone** (Support: 60%, Confidence: 100%)

Conclusion:

Thus we successfully performed market basket analysis using association rule mining using apriori algorithm.